

Jack Tadeo Murrieta Torres

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EDUCATION

Instituto Tecnológico de Sonora (ITSON)

Bachelor of Engineering in Software Engineering — Expected May 2027

Sonora, Mexico

Aug. 2023 – Present

EXPERIENCE

ACISLAB

Frontend Developer

Sonora, Mexico

Jan. 2026 – Present

- Leading the full front-end redesign of a web application for managing disability and senior care centers, improving usability and component architecture from the ground up
- Developing with **Angular**, **TypeScript**, and **RxJS**, implementing **standalone components**, **reactive forms**, **route guards**, and **lazy-loaded modules**
- Collaborating on requirements analysis, feature implementation, and UI improvements following modern front-end development practices

PROJECTS

Distributed Greenhouse Monitoring System | *Quarkus, RabbitMQ, gRPC, Docker, Vue.js* | [GitHub](#)

2026

- Designed and implemented a **distributed microservices system** with **event-driven architecture** (AMQP/RabbitMQ) for real-time sensor ingestion, alarm evaluation, and historical data storage
- Built **SensorIngestionServer** with TCP + **Adapter pattern** supporting multiple sensor brands, load-balanced across **Docker** containers with **NGINX** (Layer 4, `least_conn`)
- Integrated **gRPC** for low-latency inter-service communication, **SSE** for real-time browser notifications, and **REST/OpenAPI** for analytics and BI data exposure

CERO Dance Studio — Management System | *Java, SOLID, Gang of Four Patterns* | [GitHub](#)

2025

- Built a **real-client class and student management system** for a dance studio, covering **registration**, **scheduling**, and **event galleries**
- Applied **Factory**, **Observer**, **Strategy**, and **Decorator** patterns to solve real architectural challenges in a multi-module Java application
- Enforced **SOLID principles** throughout the codebase, reducing code duplication by approximately **50%** through systematic refactoring

EXTRACURRICULAR & VOLUNTEER ACTIVITIES

Society of Engineering Students — ITSON

Active Member

Sonora, Mexico

Aug. 2025 – Present

- Participate in academic and professional events, strengthening **teamwork**, **communication**, and **professional networking**

Construyendo Comunidades Integrales A.C.

Volunteer — Social Service

Bahía de Kino, Sonora, Mexico

Feb. 2026

- Contributed to a housing construction program that has delivered **137+ homes** to underserved families across Sonora
- Collaborated in multidisciplinary teams, developing **leadership**, **adaptability**, and **problem-solving under real constraints**

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, HTML5, CSS3

Frameworks & Tools: Angular, Quarkus, Vue.js, Spring Boot, Docker, NGINX

Databases & Messaging: MySQL, MongoDB, RabbitMQ

Concepts: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, Design Patterns, SOLID

Developer Tools: Git, GitHub, Linux, Vercel, IntelliJ IDEA, VS Code

Languages (spoken): Spanish (Native), English (B2)